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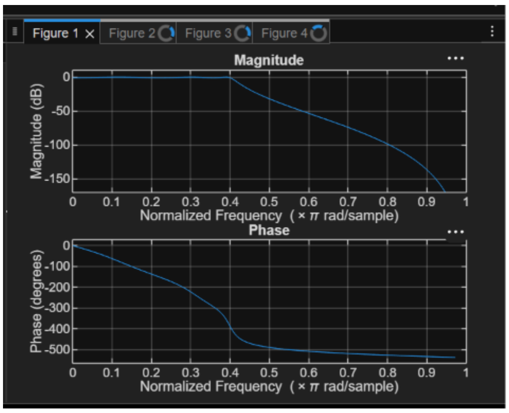
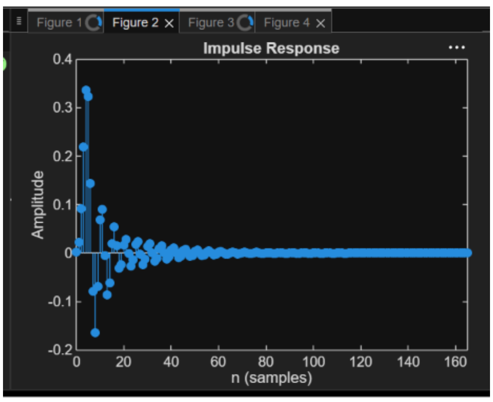
COURSE CODE: ECA0905
COURSE NAME: DIGITAL SIGNAL PROCESSING
PRACTICAL LAB RESULTS

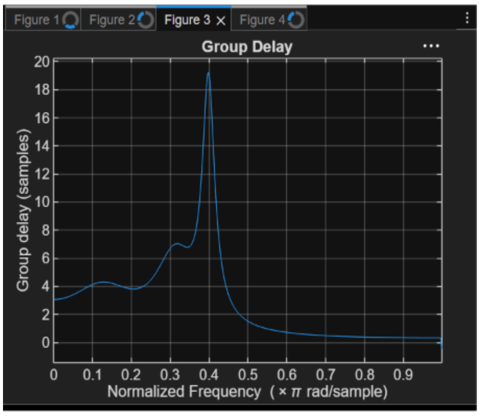
EXP: 2.1:
CODE:

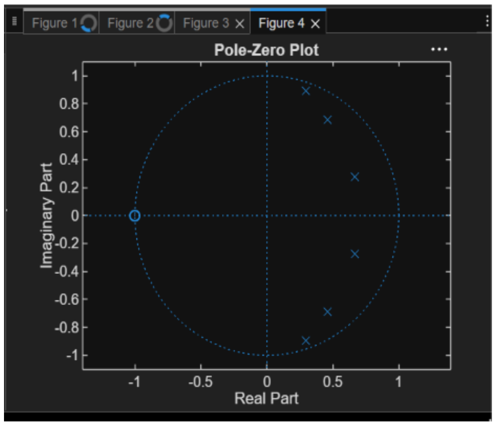
```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)

   disp('System is STABLE');

else

   disp('System is UNSTABLE');

end

% -------- GRAPHS --------

figure; freqz(b,a); % Magnitude + Phase

figure; impz(b,a); % Impulse Response

figure; grpdelay(b,a); % Group Delay

figure; zplane(b,a); % Pole-Zero
```

OUTPUT:









EXP:2.2:

CODE:

```
clc;

clear;

close all;

% Specifications

Rp = 1; % Passband Ripple (dB)

Rs = 40; % Stopband Attenuation (dB)

Wp = 0.5; % Passband Edge

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

impz(b,a);

title('Impulse Response');

% Group Delay
```

```
figure;

grpdelay( b,a);

title('Group Delay');

% Pole-Zero Plot

figure;

zplane(b,a);

title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);

fprintf('Cutoff Frequency = %.4f\n',Wn);
```

OUTPUT: